Validating final dataset

We will implement basic dataset validation checks to confirm that the final dataset is analysis-ready. This includes verifying the presence of all columns required for analysis (outcome, demographics, selected laboratory predictors, and key identifiers) and confirming that each column has the expected data type (e.g., numeric, categorical/character where applicable). We will also conduct lightweight consistency checks, such as ensuring no duplicated encounter keys, screening for implausible values (e.g., negative age or BMI).

Cohort construction and train/test split

After validating the final dataset, we will construct the modeling cohort by retaining all encounters labeled as hypoxemia and randomly sampling an equal number of non-hypoxemia encounters to form a balanced dataset for analysis. We will then split this cohort into training (70%) and test (30%) sets. The training set will be used for exploratory analyses and model development, while the test set will be reserved for final model evaluation.

Exploratory data analysis

Using the training set only, we will generate descriptive plots to visually compare the hypoxemia vs non-hypoxemia groups, including boxplots of age, BMI and key lab predictors by outcome group and bar charts comparing sex proportions across outcome groups, to compare central tendency and spread between outcome groups and detect potential outliers. If potential outliers are identified, we will assess whether they are likely data errors (e.g., unit or entry issues) or clinically plausible extreme values. Implausible values will be set to missing and handled according to our missing-data strategy, whereas plausible extreme values will be retained and addressed using appropriate transformations (e.g., log-transform for skewed variables) or winsorization (e.g., capping at percentile-based thresholds determined from the training set).

Conducting analysis

For Research Question 1: we will fit a multivariable logistic regression model on the training set with postoperative hypoxemia as the binary outcome and the full set of candidate predictors (demographics and selected laboratory predictors) as independent variables, and identify the significant predictors at the 0.05 level based on the Wald test p-values for the regression coefficients.

For Research Question 2: we will refit a reduced logistic regression model on the training set using only the significant predictors identified in Research Question 1, and evaluate predictive performance on the test set. Model performance will be evaluated using the misclassification error rate, defined as the proportion of incorrectly classified encounters.

Writing main.py script

We will implement a main.py script that serves as the main entry point for the project. Running main.py will ingest the raw .csv files, execute the full preprocessing pipeline to generate the merged dataset, run validation checks, and export the final analytic dataset. The script will then generate and save EDA figures as .png files, fit the logistic regression models, and save key outputs including a .csv file containing the estimated coefficients and Wald test p-values from the Research Question 1 (full) model, and a .txt file reporting the test-set misclassification error rate for the Research Question 2 (reduced) model.